

Module 07: Nested Subqueries and Correlated Queries

SQL Nested Subqueries vs Correlated Subqueries

Independent Evaluation vs Row-by-Row Execution, IN, ANY, ALL, EXISTS

1. Independent Nested Subqueries

Inner query runs ONCE and passes result to outer

Scalar Subquery (Single Value Return)

Uses scalar comparison operators: =, >, <, >=, <=

```
SELECT Name FROM Emp WHERE Salary > (SELECT AVG(Salary) FROM Emp);
```

Inner subquery executes once → returns scalar 50000 → outer query filters.

Multi-Row Subquery Operators:

- IN: Matches any value in returned result list.
- > ANY: Greater than at least ONE value (equivalent to > MIN).
- > ALL: Greater than EVERY value in the list (equivalent to > MAX).

```
SELECT Name FROM Emp WHERE Salary > ALL (SELECT Salary FROM Emp WHERE Dept='Sales');
```

Finds employees earning strictly more than ALL Sales employees.

Performance Characteristics:

Subquery executes exactly once. Result cached in RAM. Complexity: $O(N + M)$.

2. Correlated Subqueries & EXISTS

Inner query references outer table row & executes FOR EACH ROW

Correlated Execution Loop

```
SELECT E1.Name, E1.Salary FROM Employee E1
WHERE E1.Salary > (SELECT AVG(E2.Salary) FROM Employee E2
WHERE E2.DeptID = E1.DeptID); -- Correlation link!
```

"Find employees earning more than their own department average."
If outer table has 10,000 rows, inner subquery runs 10,000 times! $O(N*M)$.

EXISTS vs NOT EXISTS (Boolean Short-Circuit)

- EXISTS: Returns TRUE as soon as FIRST matching record is found!
- Never pulls actual data columns; only verifies existence.

```
SELECT C.CustName FROM Customers C WHERE EXISTS
(SELECT 1 FROM Orders O WHERE O.CustID = C.CustID);
```

Significantly faster than 'IN' when dealing with large subquery sets!

Golden Rule:

Use EXISTS over IN when subquery table is large and indexed!

Nested vs Correlated: Independent subqueries execute once; correlated subqueries evaluate repeatedly for every candidate row of the outer query.

1. Multi-Row Operators

Operator	Logic	Equivalence
IN	Matches any value in the returned list.	= ANY
> ANY	Greater than at least one value in the list.	> MIN(...)
> ALL	Greater than every single value in the list.	> MAX(...)

2. EXISTS Short-Circuiting

EXISTS terminates immediately upon finding the first match, making it superior to IN when querying large correlated tables.